

1 Introduction

1.1 3D Euler-Voigt Equations

The Euler-Voigt equations are an inviscid regularization of the Euler equations which have been shown to be globally well-posed and so do not form singularities given smooth initial conditions. The Euler-Voigt equations on the torus $\Omega = \mathbb{T}^3$ and time interval $[0, T]$ are defined as

$$-\alpha^2 \partial_t \Delta \mathbf{u}_\alpha + \partial_t \mathbf{u}_\alpha + \mathbf{u}_\alpha \cdot \nabla \mathbf{u}_\alpha + \nabla p = 0 \quad \text{in } \Omega \times (0, T], \quad (1.1a)$$

$$\nabla \cdot \mathbf{u}_\alpha = 0 \quad \text{in } \Omega \times (0, T], \quad (1.1b)$$

$$\mathbf{u}_\alpha(x, 0) = \boldsymbol{\eta}_\alpha. \quad (1.1c)$$

where $\mathbf{u}_\alpha$ is the velocity field, $\boldsymbol{\eta}_\alpha$ is the initial condition, and p is the scalar pressure. The regularization of the Euler-Voigt equations is accomplished with the addition of the term $-\alpha^2 \partial_t \Delta \mathbf{u}_\alpha$ where $\alpha > 0$ is the regularization parameter. As $\alpha \rightarrow 0$, the solutions to Euler-Voigt converge to the solutions of Euler. The property of constant energy, known as the α -energy, can be used to test the validity of the simulations. The α -energy is defined as

$$E_\alpha(t) := \|\mathbf{u}_\alpha(t)\|_{L^2}^2 + \alpha^2 \|\nabla \mathbf{u}_\alpha(t)\|_{L^2}^2 = E_\alpha(0). \quad (1.2)$$

We note that both Euler and Euler-Voigt possess the following time scaling property

$$\tilde{\mathbf{u}}(\mathbf{x}, t) = \lambda \mathbf{u}(\mathbf{x}, \lambda t). \quad (1.3)$$

Due to the time scaling property of the Euler-Voigt equations (1.3), the evolution of the solution will depend on the scaling of our initial condition. The initial condition we use in this analysis is the 3D Taylor Green Vortex (TGV). The TGV in the domain with side length L has the form

$$\eta_{TGV}(x) = \begin{bmatrix} V_0 \sin(2\pi x/L) \cos(2\pi y/L) \cos(2\pi z/L) \\ -V_0 \cos(2\pi x/L) \sin(2\pi y/L) \cos(2\pi z/L) \\ 0. \end{bmatrix} \quad (1.4)$$

The characteristic time-scale of (1.4) is defined as

$$t_c = \frac{L}{V_0}. \quad (1.5)$$

Our goal is to examine the behaviour of the Euler-Voigt equations as $\alpha \rightarrow 0$. In order to best compare our results to the available literature we must ensure our scaling properly matches that of previous papers which demonstrate numerical evidence for blow-up in the Euler equations from the TGV.

2 Bustamante Scale Results

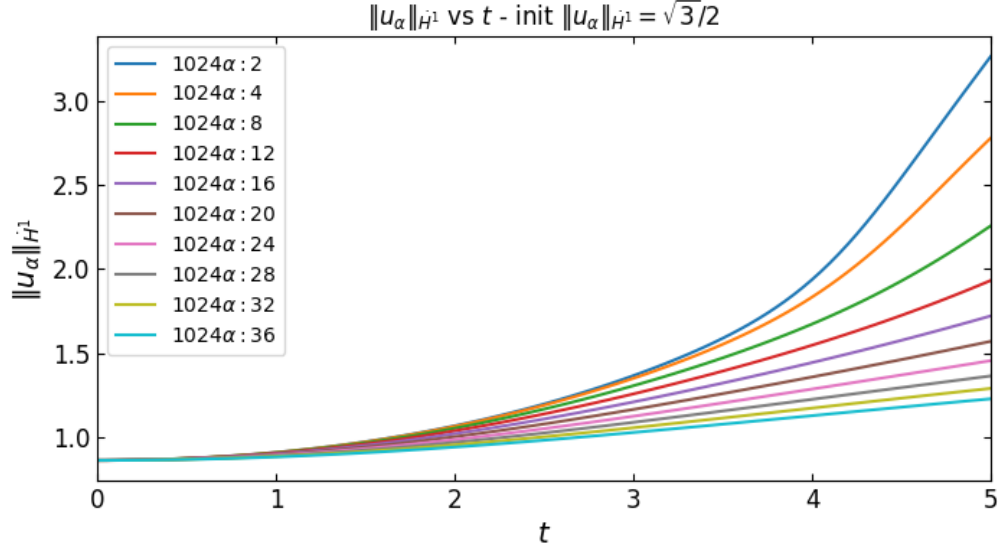
Bustamante et al. 2012 present numerical evidence for blow-up of the Euler equations from (1.4) occurring at $T^* \approx 4$. Their numerical model used parameters $L = 2\pi$ and $V_0 = 1$, and so the characteristic time-scale of their model is $t_c = 2\pi$. These parameters give their model an initial seminorm $\|\nabla \eta_{TGV}\|_{L^2} = 6\pi^3$ (A) and initial vorticity $\|\nabla \times \eta_{TGV}\|_{L^\infty} = 2$ (D).

Our numerical models are performed on a domain with $L = 1$. Thus, to match the characteristic time-scale, we use $V_0 = 1/2\pi$. These parameters give our model an initial seminorm $\|\nabla \eta_{TGV}\|_{L^2} = \sqrt{3}/2$ (C) and initial vorticity $\|\nabla \times \eta_{TGV}\|_{L^\infty} = 2$ (F).

The plots below show our results for the scaled TGV. In addition they show the comparison of our results with those of (Fig. 1b, Bustamante et al. 2012), (Fig. 2, Larios et al. 2018), and (Fig. 3, Larios et al. 2018).

2.1 Evolution of $\|\nabla \mathbf{u}_\alpha\|_{L^2}$

This plot shows the evolution of the $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ norm over time. Neither Bustamante, Brachet, or Larios show an equivalent plot but it is still worth examining. The growth in the norm strictly increases over time. Additionally, the norm at T increases as $\alpha \rightarrow 0$ which is consistent with the expected behaviour.

Figure 2.1: $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ vs t .

2.2 Evolution of $\|\omega\|_{L^\infty}$

This plot shows the evolution of the $\|\omega\|_{L^\infty}$ norm over time for each value of α . This plot is compared to Fig. 1b in Bustamante et al. 2012. Additionally, we include our reproduction of Fig 1b for Euler. The plot shows that the evolution of the vorticity for the scaled Euler-Voigt model converges to the result for the scaled Euler model, which itself matches the plot in Fig 1b. Therefore, when the initial TGV is scaled to $\|\nabla \eta_{TGV}\|_{L^2} = \sqrt{3}/2$ the solution to Euler-Voigt does match the results of Bustamante et al. 2012.

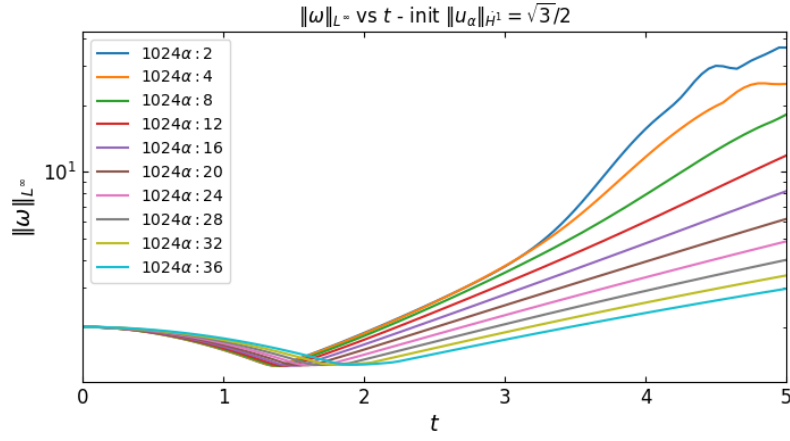
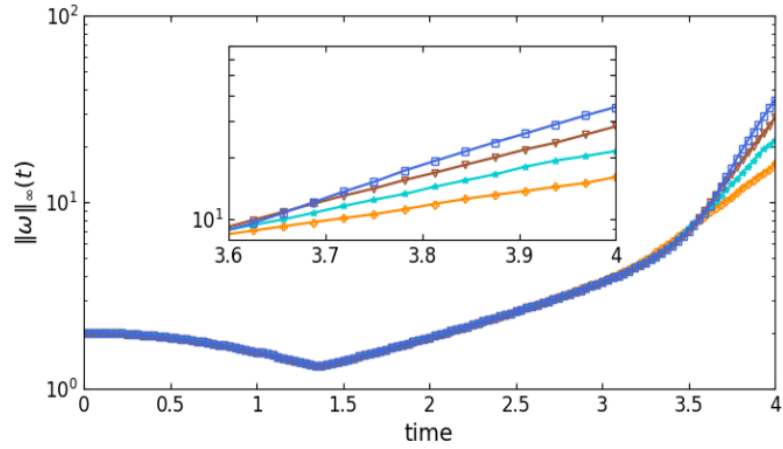
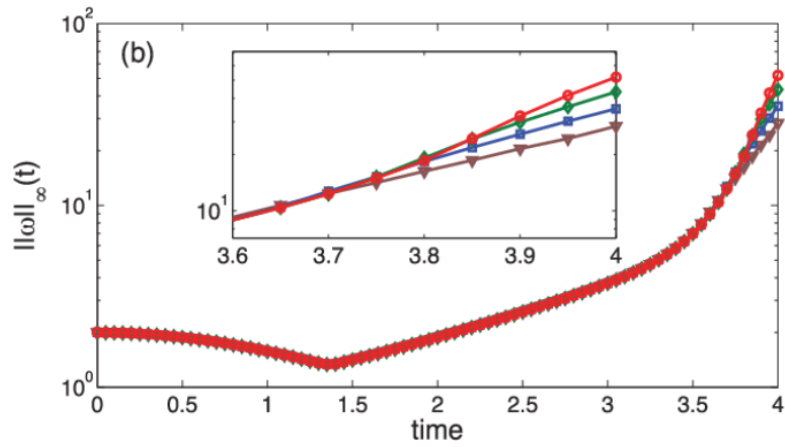
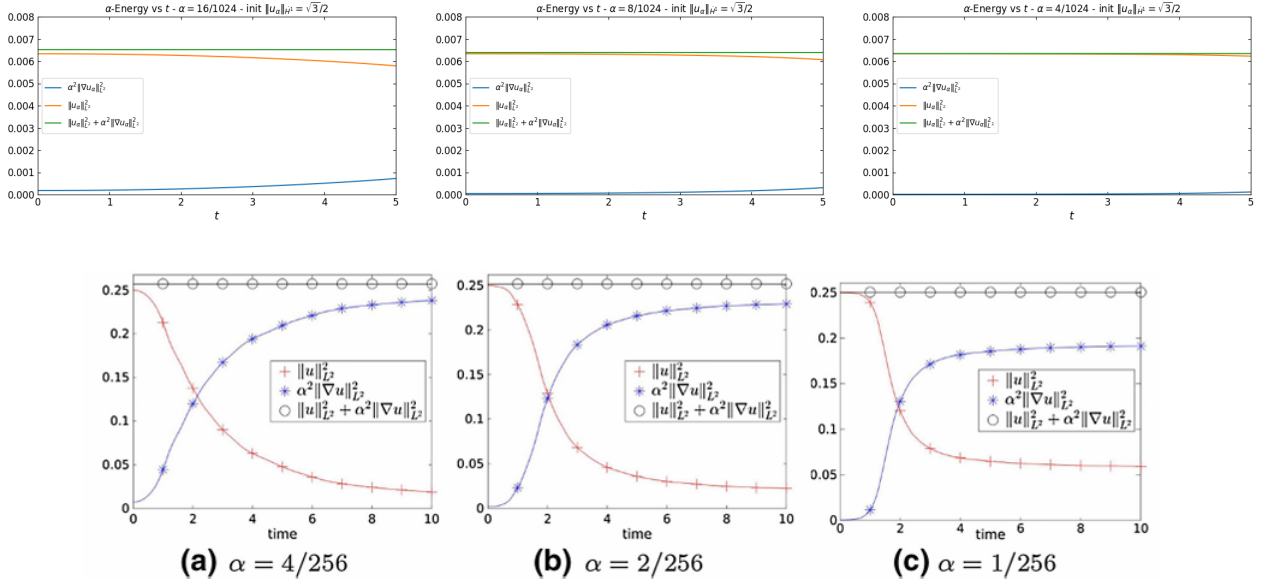

 (a) Scaled Euler-Voigt $\|\omega\|_{L^\infty}$ vs t .

 (b) Scaled Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 128, 256, 512, 1024

 (c) Bustamante Fig 1b. Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 512, 1024, 2048, 4096.

Figure 2.2: Comparison of vorticity

2.3 Reproduction of Fig 2

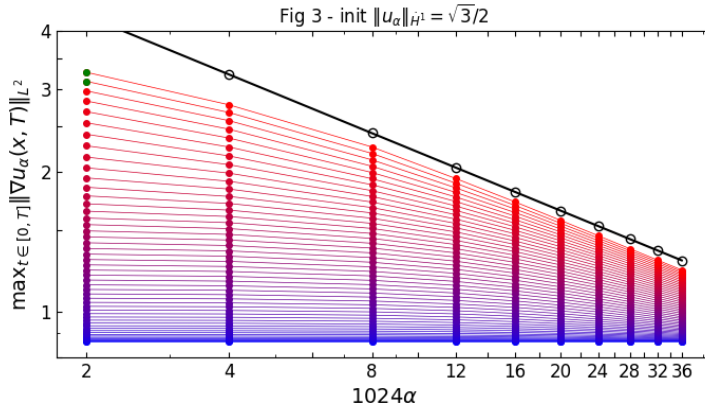
This plot shows the conservation of the α -energy for our scaled Euler-Voigt model and the conservation of α -energy from Fig. 2 of Larios et al. 2018.



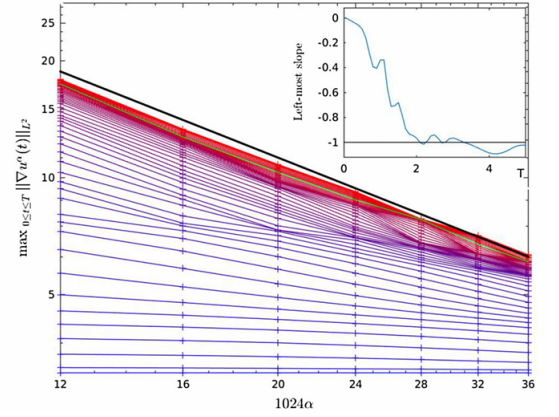
The plot shows that neither the size of the α -Energy or the time-scale it changes over match the behaviour of the model from the Larios et al. plot. However, the overall behaviour and the behaviour as $\alpha \rightarrow 0$ is consistent. The $\alpha^2 \|\nabla \mathbf{u}_\alpha\|_{L^2}^2$ term starts low and increases over time and the $\|\mathbf{u}_\alpha\|_{L^2}^2$ term starts high and decreases over time. As $\alpha \rightarrow 0$ the maximum of the $\alpha^2 \|\nabla \mathbf{u}_\alpha\|_{L^2}^2$ term decreases and the minimum of the $\|\mathbf{u}_\alpha\|_{L^2}^2$ term increases.

2.4 Reproduction of Fig 3

This plot shows our reproduction of Fig 3. from Larios et al. 2018. It shows that our results do not match. The scale of the y axis is approximately an order of magnitude smaller in our plot. Additionally the slope of the black line in the Larios plot is -1 , the slope of ours is $-.416$. Finally, our plot extends to smaller values of α . The plot shows that as $\alpha \rightarrow 0$ the slope increases which indicates that blow-up will not occur as α diminishes.



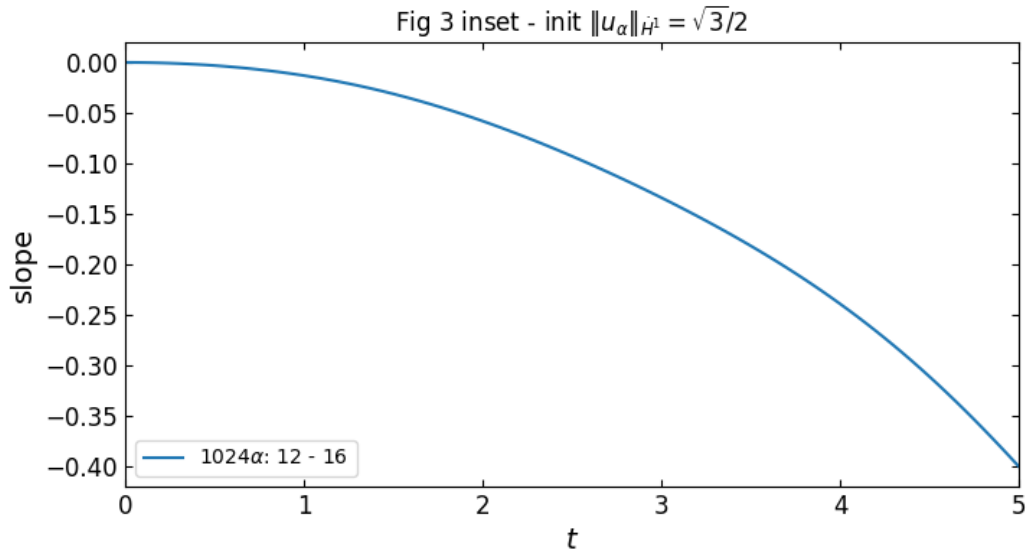
(a) Scaled Euler-Voigt. Green dots indicate the model became under-resolved.



(b) Larios et al. plot

2.5 Reproduction of Fig 3 Inset

This plot shows our reproduction of Fig 3. inset from Larios et al. 2018. It shows that our results do not match. The scale of the y axis is approximately half the size in our plot. Additionally, the minimum slope in the Larios plot is below -1 , the minimum slope of ours is $-.416$. Finally, our plot extends to smaller values of α . The plot shows that as $\alpha \rightarrow 0$ the slope increases which indicates that blow-up will not occur as α diminishes.



2.6 Conclusion

The results of our scaled Euler-Voigt model show that it matches the vorticity of the Bustamante et al. model but does not match the results of the Larios et al. model.

3 Larios Scale Results

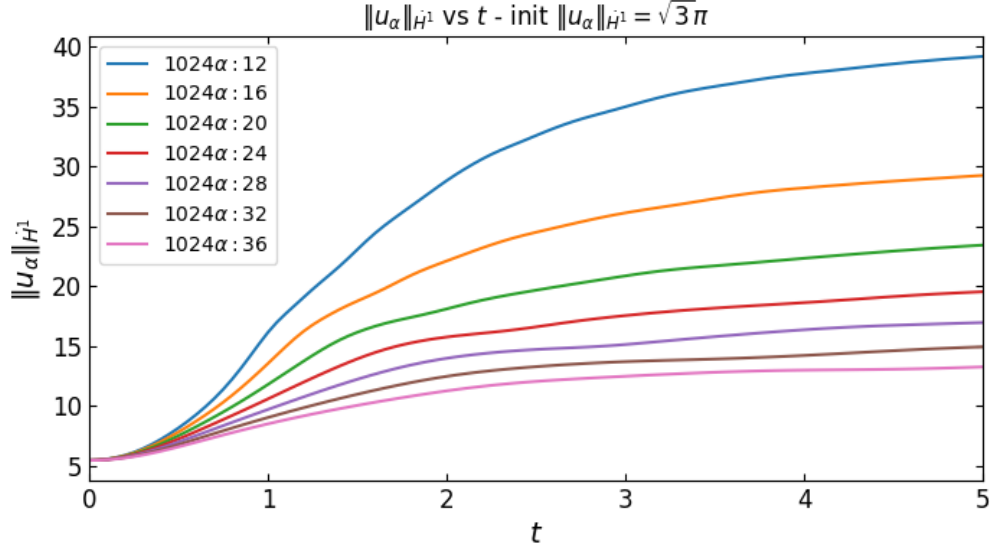
Larios et al. 2018 present numerical evidence for blow-up of the Euler equations from (1.4) occurring at $T^* \approx 4$. Their numerical model used parameters $L = 1$ and $V_0 = 1$, and so the characteristic time-scale of their model is $t_c = 1$. These parameters give their model an initial seminorm $\|\nabla \eta_{TGV}\|_{L^2} = \sqrt{3}\pi$ (B) and initial vorticity $\|\nabla \times \eta_{TGV}\|_{L^\infty} = 4\pi$ (E).

Because our model uses the same domain we do not need to scale our TGV.

The plots below show our results for the unscaled TGV. In addition they show the comparison of our results with those of (Fig. 1b, Bustamante et al. 2012), (Fig. 2, Larios et al. 2018), and (Fig. 3, Larios et al. 2018).

3.1 Evolution of $\|\nabla \mathbf{u}_\alpha\|_{L^2}$

This plot shows the evolution of the $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ norm over time. Neither Bustamante, Brachet, or Larios show an equivalent plot but it is still worth examining. The growth in the norm strictly increases over time. Additionally, the norm at T increases as $\alpha \rightarrow 0$ which is consistent with the expected behaviour. The time-scale of the results is much faster than the Bustamante scaled model. Additionally, the growth in the seminorm slows down and appears to plateau over time.

Figure 3.1: $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ vs t .

3.2 Evolution of $\|\omega\|_{L^\infty}$

This plot shows the evolution of the $\|\omega\|_{L^\infty}$ norm over time for each value of α . This plot is compared to Fig. 1b in Bustamante et al. 2012. Additionally, we include our reproduction of Fig 1b for Euler. The plot shows that the evolution of the vorticity for the unscaled Euler-Voigt model does not converge to the plot in Fig 1b. Therefore, when the initial TGV is scaled to $\|\nabla \eta_{TGV}\|_{L^2} = \sqrt{3}\pi$ the solution to Euler-Voigt does not match the results of Bustamante et al. 2012.

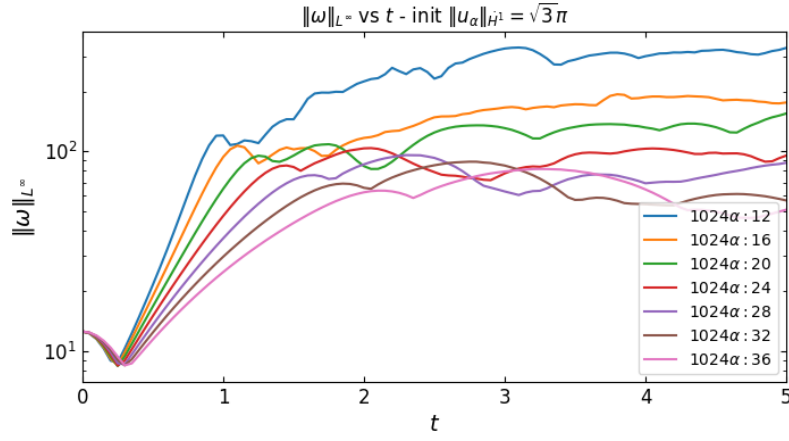
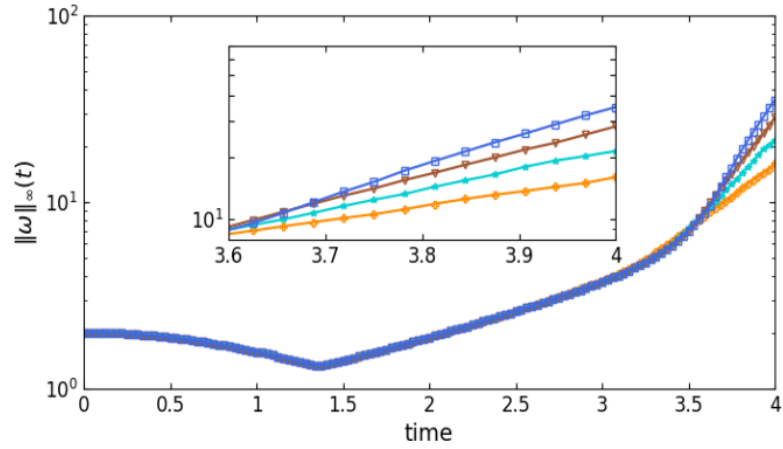
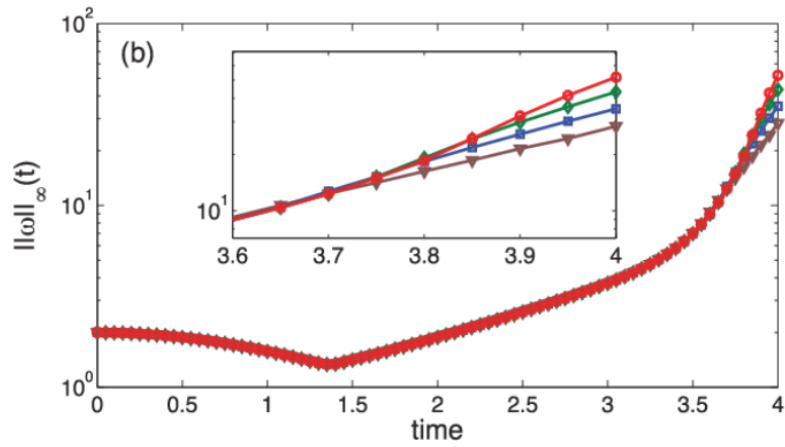
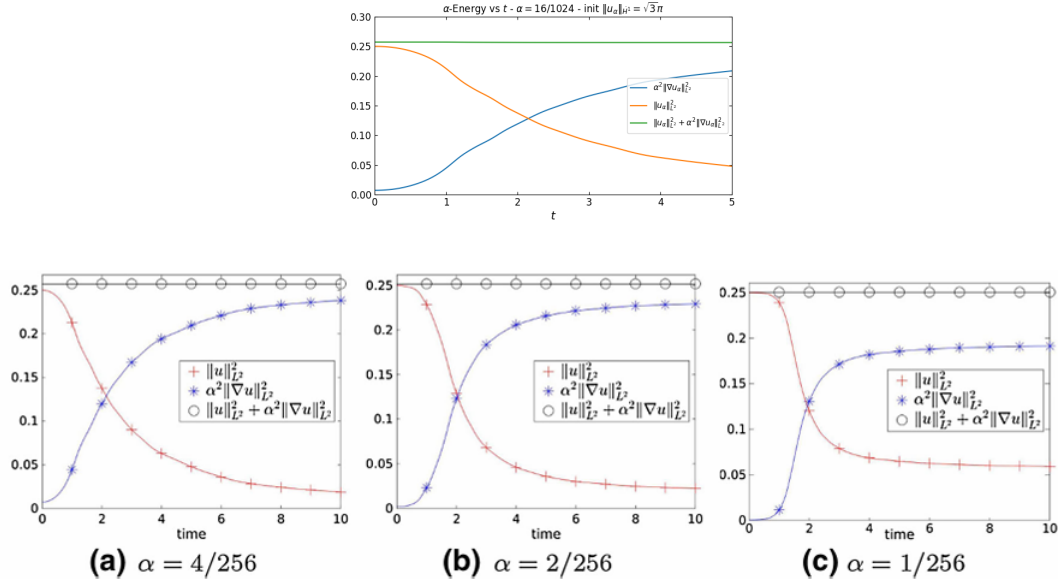
(a) Scaled Euler-Voigt $\|\omega\|_{L^\infty}$ vs t .(b) Scaled Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 128, 256, 512, 1024(c) Bustamante Fig 1b. Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 512, 1024, 2048, 4096.

Figure 3.2: Comparison of vorticity

3.3 Reproduction of Fig 2

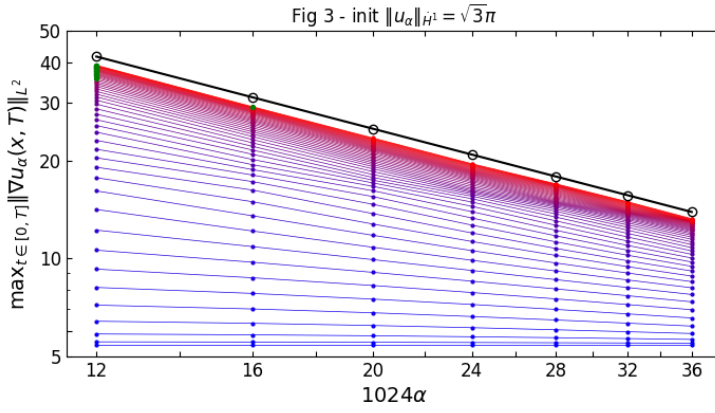
This plot shows the conservation of the α -energy for the Euler-Voigt model and the conservation of α -energy from Fig. 2 of Larios et al. 2018.



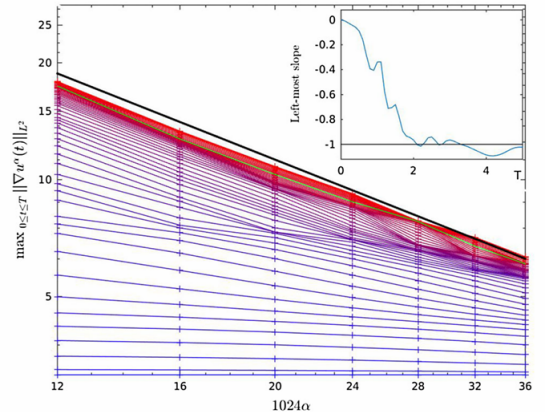
The plot shows that both the size of the α -Energy and the time-scale it changes over match the behaviour of the model from the Larios et al. plot. Plots for $\alpha = 2/256$ and $\alpha = 1/256$ are not included because they became under-resolved.

3.4 Reproduction of Fig 3

This plot shows our reproduction of Fig 3. from Larios et al. 2018. It shows that our results partly match. The scale of the y axis is approximately twice as large in our plot. The slope of the black line in the Larios plot is -1 , the slope of ours is also -1 , indicating that the behaviour as $\alpha \rightarrow 0$ matches.



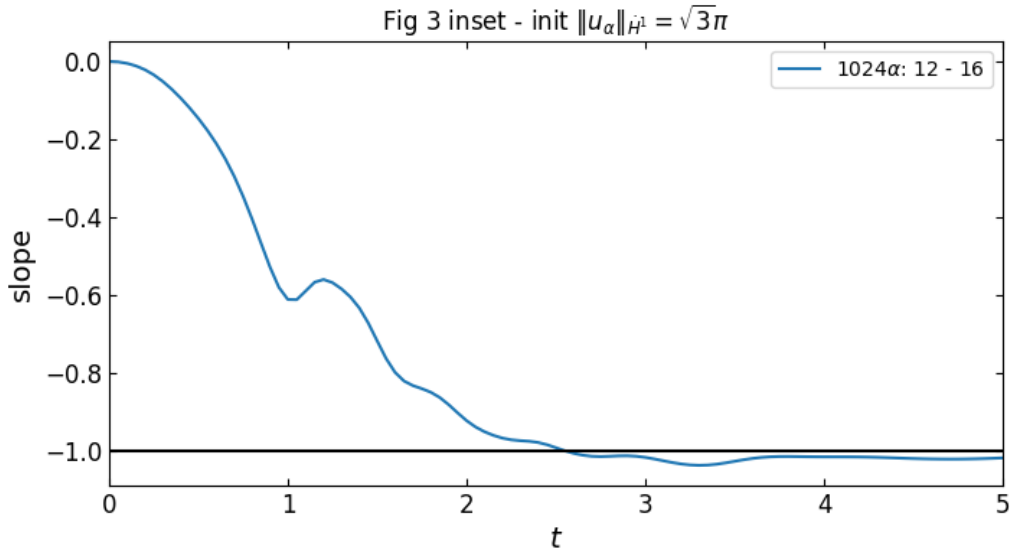
(a) Scaled Euler-Voigt. Green dots indicate the model became under-resolved.



(b) Larios et al. plot

3.5 Reproduction of Fig 3 Inset

This plot shows our reproduction of Fig 3. inset from Larios et al. 2018. It shows that our results approximately match. The scale of the y axis is the same and that both the minimum slope in the Larios plot and of ours is below -1 . However, the minimums do not occur at the same time as ours occurs earlier at $T^* \approx 3.2$.



3.6 Conclusion

The results of our scaled Euler-Voigt model show that it does not match the time-scale of the Bustamante et al. model, it does match the Larios et al. model for Fig 2 but not for Fig 3.

4 Larios Scale 2 Results

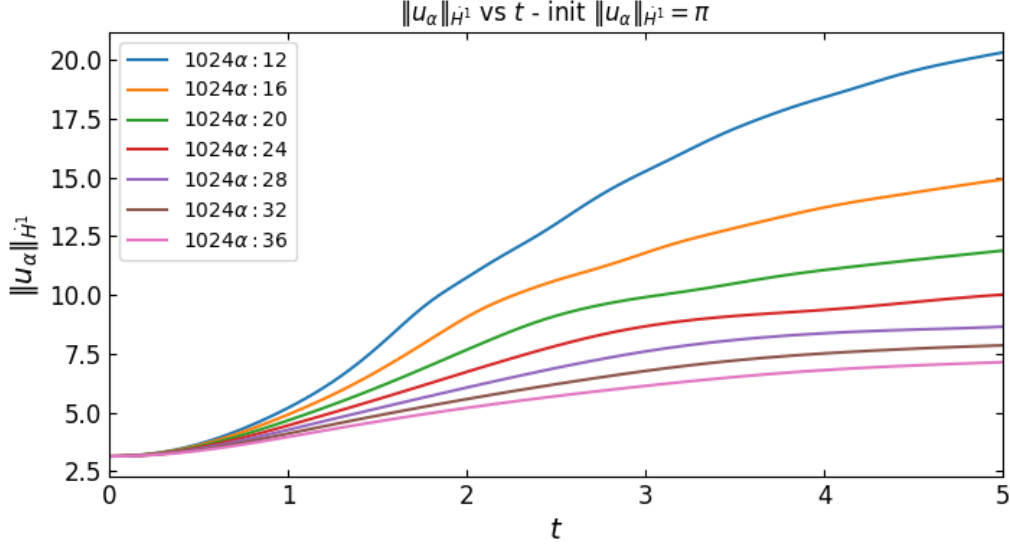
Larios et al. 2018 present numerical evidence for blow-up of the Euler-Voigt equations from (1.4) occurring at $T^* \approx 4$. Their numerical model used parameters $L = 1$ and $V_0 = 1$, and so the characteristic time-scale of their model is $t_c = 1$. These parameters give their model an initial seminorm $\|\nabla \eta_{TGV}\|_{L^2} = \sqrt{3}\pi$ (B) and initial vorticity $\|\nabla \times \eta_{TGV}\|_{L^\infty} = 4\pi$ (E).

Our numerical models are performed on a domain with the same size and so does not need to be rescaled. However, at this scaling the results of our model do not match the results of Fig 3. Therefore, we introduce an ad hoc scaling to $\|\nabla \eta_{TGV}\|_{L^2} = \pi$.

The plots below show our results for the scaled TGV. In addition they show the comparison of our results with those of (Fig. 1b, Bustamante et al. 2012), (Fig. 2, Larios et al. 2018), and (Fig. 3, Larios et al. 2018).

4.1 Evolution of $\|\nabla \mathbf{u}_\alpha\|_{L^2}$

This plot shows the evolution of the $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ norm over time. Neither Bustamante, Brachet, or Larios show an equivalent plot but it is still worth examining. The growth in the norm strictly increases over time. Additionally, the norm at T increases as $\alpha \rightarrow 0$ which is consistent with the expected behaviour. The time-scale of the results is between the Bustamante scaled model and the first Larios scaling.

Figure 4.1: $\|\nabla \mathbf{u}_\alpha\|_{L^2}$ vs t .

4.2 Evolution of $\|\omega\|_{L^\infty}$

This plot shows the evolution of the $\|\omega\|_{L^\infty}$ norm over time for each value of α . This plot is compared to Fig. 1b in Bustamante et al. 2012. Additionally, we include our reproduction of Fig 1b for Euler. The plot shows that the evolution of the vorticity for the second scaled Euler-Voigt model does not converge to the plot in Fig 1b. Therefore, when the initial TGV is scaled to $\|\nabla \eta_{TGV}\|_{L^2} = \pi$ the solution to Euler-Voigt does not match the results of Bustamante et al. 2012.

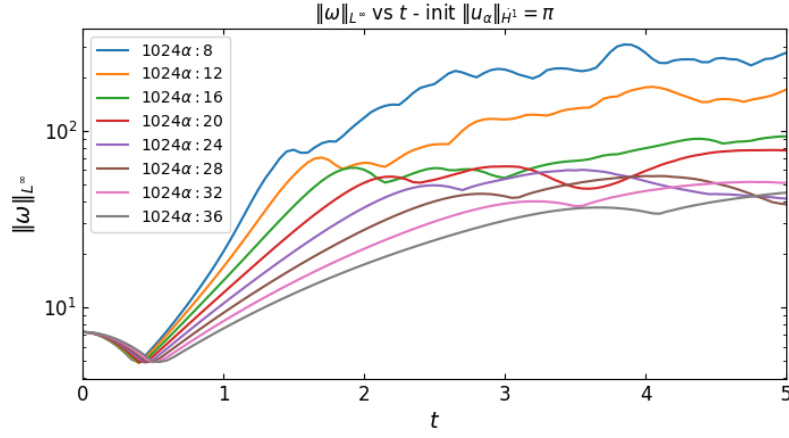
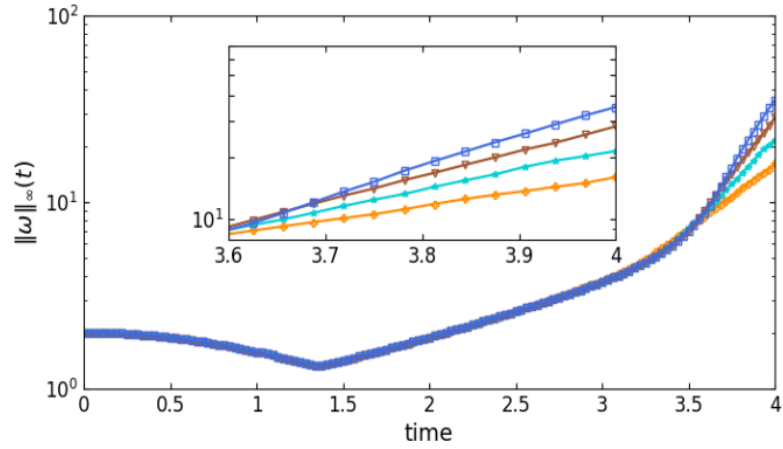
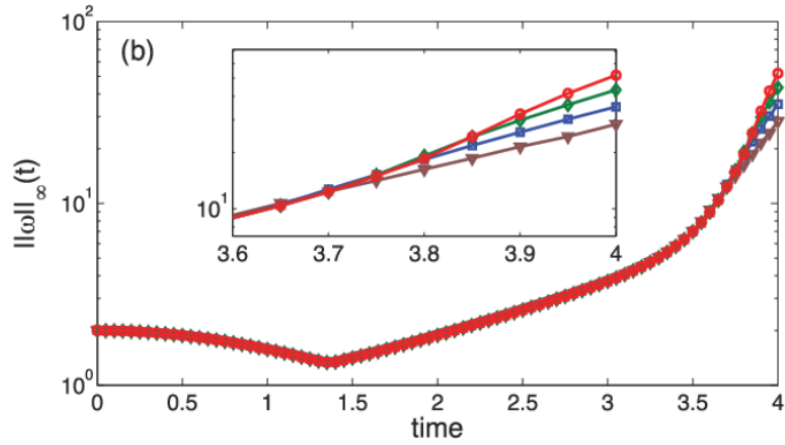
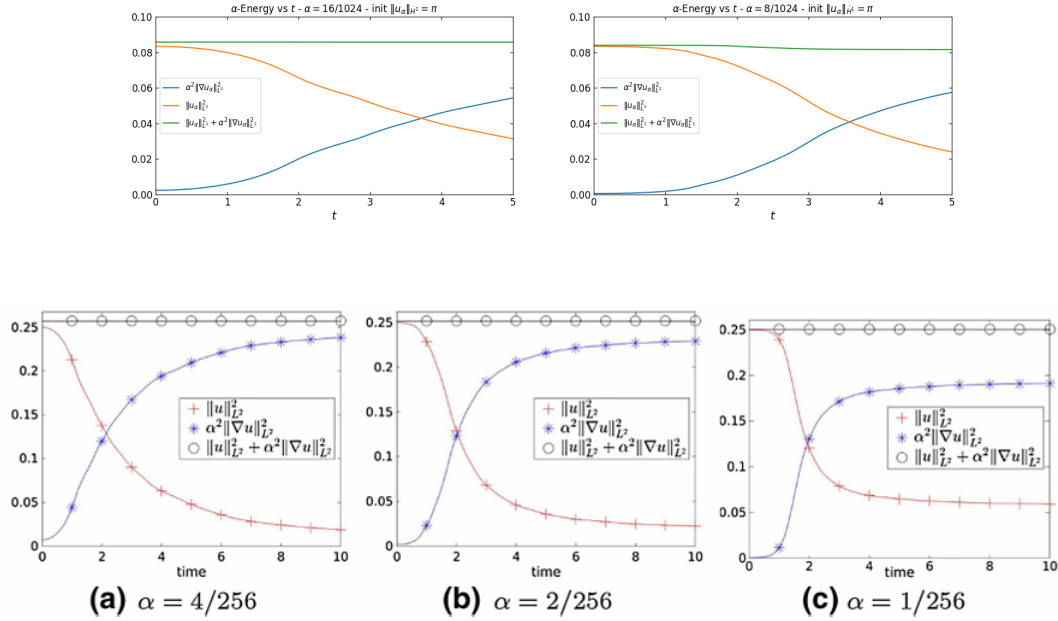
(a) Scaled Euler-Voigt $\|\omega\|_{L^\infty}$ vs t .(b) Scaled Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 128, 256, 512, 1024(c) Bustamante Fig 1b. Euler $\|\omega\|_{L^\infty}$ vs t . Resolutions: 512, 1024, 2048, 4096.

Figure 4.2: Comparison of vorticity

4.3 Reproduction of Fig 2

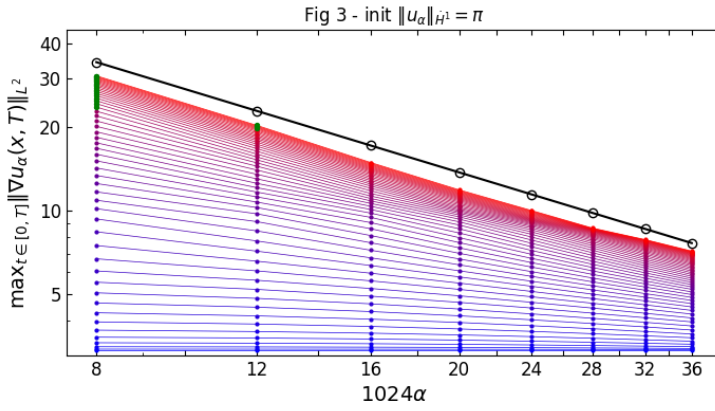
This plot shows the conservation of the α -energy for the Euler-Voigt model and the conservation of α -energy from Fig. 2 of Larios et al. 2018.



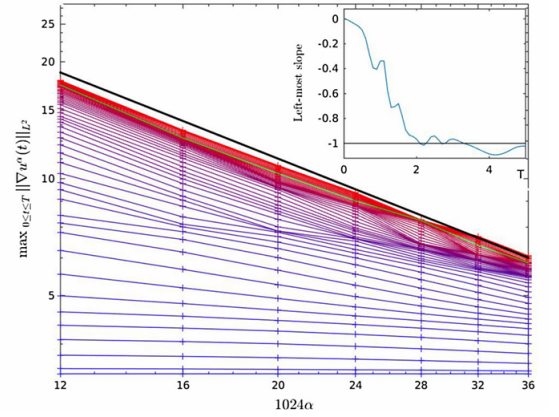
The plot shows that both the size of the α -Energy and the time-scale it changes over do not match the behaviour of the model from the Larios et al. plot. The plot for $\alpha = 1/256$ is not included because it became under-resolved.

4.4 Reproduction of Fig 3

This plot shows our reproduction of Fig 3. from Larios et al. 2018. It shows that our results match. The scale of the y axis is the same as in our plot. The slope of the black line in the Larios plot is -1 , the slope of ours is also -1 .



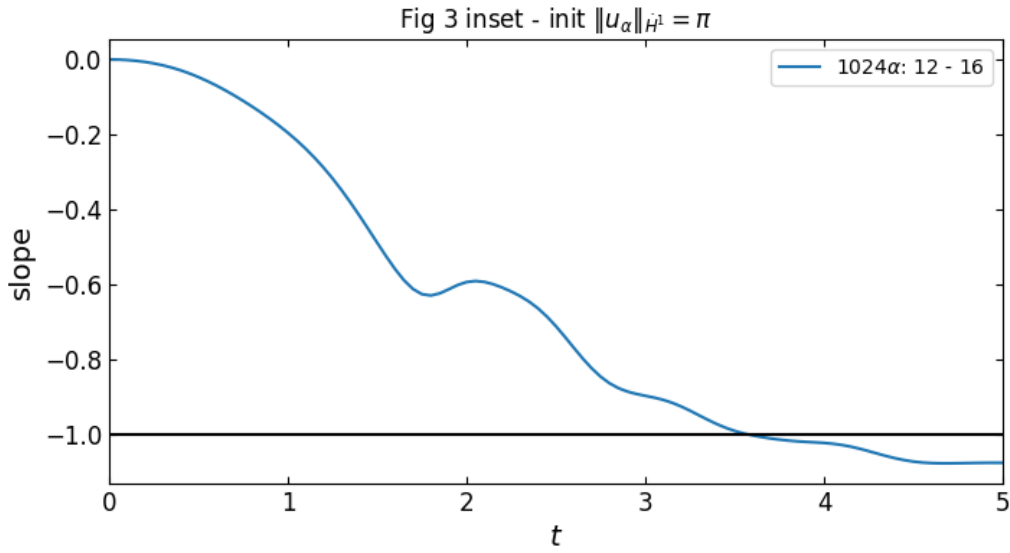
(a) Scaled Euler-Voigt. Green dots indicate the model became under-resolved.



(b) Larios et al. plot

4.5 Reproduction of Fig 3 Inset

This plot shows our reproduction of Fig 3. inset from Larios et al. 2018. It shows that our results match. The scale of the y axis is the same. The minimum slope in the Larios plot is below -1 , the minimum slope of ours is also below -1 . Finally, the time at which the slope reaches its minimum value is approximately the same as the Larios model.



4.6 Conclusion

The results of our second scaled Euler-Voigt model show that it does not match the time-scale of the Bustamante et al. model, or the results of Fig 2, but does match the results of Larios et al. Fig 3.

A Taylor Green $\dot{H}^1$ in $[0, 2\pi]^3$ domain

The 3D Taylor Green initial condition on the domain $[0, 2\pi]^3$ is

$$\eta_{TG} = \begin{bmatrix} \sin(x) \cos(y) \cos(z) \\ -\cos(x) \sin(y) \cos(z) \\ 0. \end{bmatrix}$$

The formula for the H^1 seminorm is

$$\|\eta\|_{\dot{H}^1} = \|\nabla \eta\|_{L^2} = \left[\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx \right]^{1/2}.$$

Because the z component of η_{TG} is zero, the partial derivative is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (\partial_x \eta_x)^2 dx + \int_{\Omega} (\partial_y \eta_x)^2 dx \\ &\quad + \int_{\Omega} (\partial_z \eta_x)^2 dx + \int_{\Omega} (\partial_x \eta_y)^2 dx \\ &\quad + \int_{\Omega} (\partial_y \eta_y)^2 dx + \int_{\Omega} (\partial_z \eta_y)^2 dx. \end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (\cos(x) \cos(y) \cos(z))^2 dx + \int_{\Omega} (-\sin(x) \sin(y) \cos(z))^2 dx \\ &\quad + \int_{\Omega} (-\sin(x) \cos(y) \sin(z))^2 dx + \int_{\Omega} (\sin(x) \sin(y) \cos(z))^2 dx \\ &\quad + \int_{\Omega} (-\cos(x) \cos(y) \cos(z))^2 dx + \int_{\Omega} (\cos(x) \sin(y) \sin(z))^2 dx. \end{aligned}$$

Evaluating each integral

$$\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx = \pi^3 + \pi^3 + \pi^3 + \pi^3 + \pi^3 + \pi^3 = 6\pi^3.$$

Substituting into the formula for the H^1 seminorm

$$\|\eta\|_{\dot{H}^1} = [6\pi^3]^{1/2}.$$

B Taylor Green $\dot{H}^1$ in $[0, 1]^3$ domain

The 3D Taylor Green initial condition on the domain $[0, 1]^3$ is

$$\eta_{TG} = \begin{bmatrix} \sin(2\pi x) \cos(2\pi y) \cos(2\pi z) \\ -\cos(2\pi x) \sin(2\pi y) \cos(2\pi z) \\ 0. \end{bmatrix}$$

The formula for the H^1 seminorm is

$$\|\eta\|_{\dot{H}^1} = \|\nabla \eta\|_{L^2} = \left[\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx \right]^{1/2}.$$

Because the z component of η_{TG} is zero, the partial derivative is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (\partial_x \eta_x)^2 dx + \int_{\Omega} (\partial_y \eta_x)^2 dx \\ &\quad + \int_{\Omega} (\partial_z \eta_x)^2 dx + \int_{\Omega} (\partial_x \eta_y)^2 dx \\ &\quad + \int_{\Omega} (\partial_y \eta_y)^2 dx + \int_{\Omega} (\partial_z \eta_y)^2 dx. \end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned}
\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (2\pi \cos(2\pi x) \cos(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-2\pi \sin(2\pi x) \sin(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-2\pi \sin(2\pi x) \cos(2\pi y) \sin(2\pi z))^2 dx \\
&+ \int_{\Omega} (2\pi \sin(2\pi x) \sin(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-2\pi \cos(2\pi x) \cos(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (2\pi \cos(2\pi x) \sin(2\pi y) \sin(2\pi z))^2 dx.
\end{aligned}$$

Evaluating each integral

$$\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx = \frac{\pi^2}{2} + \frac{\pi^2}{2} + \frac{\pi^2}{2} + \frac{\pi^2}{2} + \frac{\pi^2}{2} + \frac{\pi^2}{2} = 3\pi^2.$$

Substituting into the formula for the H^1 seminorm

$$\|\eta\|_{\dot{H}^1} = [3\pi^2]^{1/2} = \sqrt{3}\pi.$$

C Scaled Taylor Green $\dot{H}^1$ in $[0, 1]^3$ domain

The scaled 3D Taylor Green initial condition on the domain $[0, 1]^3$ is

$$\eta_{TG} = \begin{bmatrix} (1/2\pi) \sin(2\pi x) \cos(2\pi y) \cos(2\pi z) \\ -(1/2\pi) \cos(2\pi x) \sin(2\pi y) \cos(2\pi z) \\ 0. \end{bmatrix}$$

The formula for the H^1 seminorm is

$$\|\eta\|_{\dot{H}^1} = \|\nabla \eta\|_{L^2} = \left[\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx \right]^{1/2}.$$

Because the z component of η_{TG} is zero, the partial derivative is also zero. This leaves us with six integrals to evaluate.

$$\begin{aligned} \sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (\partial_x \eta_x)^2 dx + \int_{\Omega} (\partial_y \eta_x)^2 dx \\ &\quad + \int_{\Omega} (\partial_z \eta_x)^2 dx + \int_{\Omega} (\partial_x \eta_y)^2 dx \\ &\quad + \int_{\Omega} (\partial_y \eta_y)^2 dx + \int_{\Omega} (\partial_z \eta_y)^2 dx. \end{aligned}$$

Evaluating each partial derivative

$$\begin{aligned}
\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx &= \int_{\Omega} (\cos(2\pi x) \cos(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-\sin(2\pi x) \sin(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-\sin(2\pi x) \cos(2\pi y) \sin(2\pi z))^2 dx \\
&+ \int_{\Omega} (\sin(2\pi x) \sin(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (-\cos(2\pi x) \cos(2\pi y) \cos(2\pi z))^2 dx \\
&+ \int_{\Omega} (\cos(2\pi x) \sin(2\pi y) \sin(2\pi z))^2 dx.
\end{aligned}$$

Evaluating each integral

$$\sum_{i,j=1}^3 \int_{\Omega} (\partial_i \eta_j)^2 dx = \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8} = \frac{3}{4}$$

Substituting into the formula for the H^1 seminorm

$$\|\eta\|_{\dot{H}^1} = \left[\frac{3}{4} \right]^{1/2} = \frac{\sqrt{3}}{2}.$$

D Taylor Green $\|\nabla \times \eta\|_{L^\infty}$ in $[0, 2\pi]^3$ domain

The 3D Taylor Green initial condition on the domain $[0, 2\pi]^3$ is

$$\eta_{TG} = \begin{bmatrix} \sin(x) \cos(y) \cos(z) \\ -\cos(x) \sin(y) \cos(z) \\ 0. \end{bmatrix}$$

The vorticity of η_{TG} is

$$\nabla \times \eta_{TG} = \begin{bmatrix} \partial_y \eta_z - \partial_z \eta_y \\ \partial_z \eta_x - \partial_x \eta_z \\ \partial_x \eta_y - \partial_y \eta_x. \end{bmatrix}$$

Substituting in the partial derivatives

$$\nabla \times \eta_{TG} = \begin{bmatrix} 0 - \cos(x) \sin(y) \sin(z) \\ -\sin(x) \cos(y) \sin(z) - 0 \\ \sin(x) \sin(y) \cos(z) + \sin(x) \sin(y) \cos(z). \end{bmatrix}$$

which simplifies to

$$\nabla \times \eta_{TG} = \begin{bmatrix} -\cos(x) \sin(y) \sin(z) \\ -\sin(x) \cos(y) \sin(z) \\ 2 \sin(x) \sin(y) \cos(z). \end{bmatrix}$$

The $\|\nabla \times \eta_{TG}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \eta_{TG}$. Therefore,

$$\|\nabla \times \eta_{TG}\|_{L^\infty} = 2$$

E Taylor Green $\|\nabla \times \eta\|_{L^\infty}$ in $[0, 1]^3$ domain

The 3D Taylor Green initial condition on the domain $[0, 1]^3$ is

$$\eta_{TG} = \begin{bmatrix} \sin(2\pi x) \cos(2\pi y) \cos(2\pi z) \\ -\cos(2\pi x) \sin(2\pi y) \cos(2\pi z) \\ 0. \end{bmatrix}$$

The vorticity of η_{TG} is

$$\nabla \times \eta_{TG} = \begin{bmatrix} \partial_y \eta_z - \partial_z \eta_y \\ \partial_z \eta_x - \partial_x \eta_z \\ \partial_x \eta_y - \partial_y \eta_x. \end{bmatrix}$$

Substituting in the partial derivatives

$$\nabla \times \eta_{TG} = \begin{bmatrix} 0 - 2\pi \cos(2\pi x) \sin(2\pi y) \sin(2\pi z) \\ -2\pi \sin(2\pi x) \cos(2\pi y) \sin(2\pi z) - 0 \\ 2\pi \sin(2\pi x) \sin(2\pi y) \cos(2\pi z) + 2\pi \sin(2\pi x) \sin(2\pi y) \cos(2\pi z). \end{bmatrix}$$

which simplifies to

$$\nabla \times \eta_{TG} = \begin{bmatrix} -2\pi \cos(2\pi x) \sin(2\pi y) \sin(2\pi z) \\ -2\pi \sin(2\pi x) \cos(2\pi y) \sin(2\pi z) \\ 4\pi \sin(2\pi x) \sin(2\pi y) \cos(2\pi z). \end{bmatrix}$$

The $\|\nabla \times \eta_{TG}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \eta_{TG}$. Therefore,

$$\|\nabla \times \eta_{TG}\|_{L^\infty} = 4\pi$$

F Scaled Taylor Green $\|\nabla \times \eta\|_{L^\infty}$ in $[0, 1]^3$ domain

The 3D Taylor Green initial condition on the domain $[0, 1]^3$ is

$$\eta_{TG} = \begin{bmatrix} (1/2\pi) \sin(2\pi x) \cos(2\pi y) \cos(2\pi z) \\ -(1/2\pi) \cos(2\pi x) \sin(2\pi y) \cos(2\pi z) \\ 0. \end{bmatrix}$$

The vorticity of η_{TG} is

$$\nabla \times \eta_{TG} = \begin{bmatrix} \partial_y \eta_z - \partial_z \eta_y \\ \partial_z \eta_x - \partial_x \eta_z \\ \partial_x \eta_y - \partial_y \eta_x. \end{bmatrix}$$

Substituting in the partial derivatives

$$\nabla \times \eta_{TG} = \begin{bmatrix} 0 - \cos(2\pi x) \sin(2\pi y) \sin(2\pi z) \\ -\sin(2\pi x) \cos(2\pi y) \sin(2\pi z) - 0 \\ \sin(2\pi x) \sin(2\pi y) \cos(2\pi z) + \sin(2\pi x) \sin(2\pi y) \cos(2\pi z). \end{bmatrix}$$

which simplifies to

$$\nabla \times \eta_{TG} = \begin{bmatrix} -\cos(2\pi x) \sin(2\pi y) \sin(2\pi z) \\ -\sin(2\pi x) \cos(2\pi y) \sin(2\pi z) \\ 2 \sin(2\pi x) \sin(2\pi y) \cos(2\pi z). \end{bmatrix}$$

The $\|\nabla \times \eta_{TG}\|_{L^\infty}$ norm is the maximum absolute value of the components of $\nabla \times \eta_{TG}$. Therefore,

$$\|\nabla \times \eta_{TG}\|_{L^\infty} = 2$$